

How Observation Error is Added to a Simulated Composition Data Matrix $\mathbf{X}$

In the hake.R script, the generation of the composition count data (row g in $\mathbf{X}$) for each fleet group g involves a two-stage Dirichlet-Multinomial process with added log-normal noise. Here is the step-by-step breakdown:

1. **Proportion Perturbation (Observation Error):** The true simplex proportion (p_{true}) is transformed into log-space, where random normal noise with a mean of 0 and a standard deviation of σ is added ($z \leftarrow \log(p_{\text{true}}) + \text{rnorm}(K, 0, \sigma)$). It is then exponentiated and normalized to sum to 1 to create a group-specific proportion (p_g).
2. **Group-Specific Theta:** A group-specific concentration multiplier (θ_g) is drawn from a lognormal distribution. The mean of this distribution is based on θ_{true} , and its variance is derived from θ_{CV} .
3. **Overdispersion Parameter Calculation:** The Dirichlet precision/concentration parameter (a_0g) is calculated as the product of the group-specific θ , the overdispersion multiplier, and the simulated sample size ($a_0g \leftarrow (\theta_g * \text{od_mult}) * N_{\text{row}}[g]$).
4. **Gamma-Multinomial Sampling (Dirichlet equivalent):** The shape parameters for a Gamma distribution (α_{vec}) are calculated as $a_0g * p_g$. Random probabilities (ϕ) are drawn from this Gamma distribution and normalized. Finally, the count data for the group ($X[g,]$) is drawn from a Multinomial distribution using the sample size ($N_{\text{row}}[g]$) and the perturbed probabilities (ϕ).

Input Parameters Affecting Simulated Observation Error and Sample Size

Table 1. This table summarizes how specific parameters from the input file hake.inp affect observation error, overdispersion, and simulated sample size by fleet group.

Parameter	Value	Effect on Observation Error	Effect on Overdispersion	Effect on Sample Size
theta_true	10	None directly.	Decreases overdispersion. Acts as the baseline precision scale. Higher values increase $a0g$, making the drawn probabilities (ϕ) cluster more tightly around p_g .	None.
theta_CV	0.2	None directly.	Increases variability of overdispersion. Adds group-to-group lognormal variance to θeta_g , meaning some groups will be more overdispersed than others.	None.
od_mult	0.5	None directly.	Increases overdispersion. Directly scales down the precision parameter $a0g$. Values less than 1 spread the Dirichlet distribution wider.	None.
sigma	0.5	Primary driver of observation error. Adds random noise to the true proportions in log-space before they are used to generate the count data.	Alters the <i>mean</i> around which the Dirichlet distribution is centered, but doesn't dictate the spread (dispersion) itself.	None.
mean_nsamp	100	None directly.	Decreases overdispersion. Because $a0g$ is multiplied by the sample size, larger average sample sizes increases Dirichlet precision.	Sets the target mean for the sample size distribution.
ln_sd	1.0	None directly.	Indirectly impacts overdispersion by creating wider variance in group sample sizes ($N_row[g]$).	Determines the standard deviation of lognormal sample sizes.